

Capstone Project: QuickAid – Smart Campus Helpdesk

Problem Statement

You are a team of Azure developers and have been tasked by a university (your client) to create an application called “QuickAid – Smart Campus Helpdesk” for students and staff to submit issues, track requests, and receive timely support on campus. In a very competitive EdTech market, you want to ensure your application is selected and potentially scaled to other institutions.

Work together as a team to create the application using all the technical content knowledge and soft skills you have gained from the MyMahir Microsoft Developer programme and CE21’s 3-module digital courseware.

Present your final product via slides showing your development process with a live demo on how your product works. Each team member covers one stage (research, design, implementation, testing, management, evaluation). Your presentation should be structured into: Intro, Body (stages in the Excel document), Evaluation, Conclusion.

Assessment covers technical work and soft skills (collaboration, communication, etc.). Each student will be awarded a competency Score card based on an assessment of:

- all recorded team discussions
- all individual work submissions
- all mind-map scores
- individual project management excel sheet
- final product presentation and demo plus collated project management excel doc used in the presentation

You can **create a portfolio** to show employers the product, the competency badges you receive, your score card, and a copy of your project management skills document, after you complete the course.

Project Requirements for Application Developed

Functional Requirements:

- Web form interface for submitting helpdesk tickets
- Endpoint for retrieving tickets based on user email
- Automated email notifications upon ticket submission
- Persistent ticket storage with unique identifiers and status tracking

Technical Requirements:

- Frontend hosted on Azure App Service
- Backend powered by Azure Functions using Python
- Ticket data stored in Azure Cosmos DB
- Secret management using Azure Key Vault
- Email integration through SendGrid
- All components must be deployable under the Azure Free Tier

Learning Objectives

- Gain hands-on experience with Azure App Services, Functions, and Cosmos DB
- Understand the principles of serverless and event-driven application development
- Learn to integrate external services such as SendGrid for communication features
- Apply best practices in security by using Azure Key Vault for secret management
- Develop and deploy a cloud-native application that is scalable, maintainable, and showcase-ready

Project Requirements for Presentation

Presentation Structure

- Start off with slide deck showing work distribution, system diagram and screenshots of system. Please highlight any additional features your added on
- Present working system
- Documentation of system deployments and setup

Rubric: What categories will students be assessed on. Describe what you expect to see and how you will differentiate the competency levels

Assessment Rubric

Total Marks: 105 (including 5 bonus marks for Application Insights)

Criteria	Marks
Deployment (Frontend + Backend)	20
Working Features (Submit & Retrieve Tickets)	20
Integration of SendGrid Email Notifications	15
Secure Secret Management via Azure Key Vault	15

Documentation (README, Architecture Diagram, Setup Steps)	10
Customization or Feature Enhancement	10
Team Collaboration (Commits, Roles, Division of Work)	10
Optional: Application Insights Integration	5

Assessment Rubric for Presentation (20 min presentation 10 min Q&A)

Presentation skills (voice, pace and speed of delivery, eye contact, preparedness, language accuracy, timing)

Content & flow of information: robust content, well connected ideas, logical reasoning and flow of ideas from one slide to the next drawn to a sound conclusion (or implications, if any)

Visuals and design: appearance and effectiveness of slide design and use of animations to support the communication of content

Rubric		ThinkPeak			
		Excellent (80-100%)	Good (70-79)	Improvement Required (60-69)	Not Acceptable (Below 60)
Content & Flow :					
quality of ideas					
logical flow of ideas (cohesion between slides)					
logical reasoning					
Visuals:					
appearance (font size, colour, design, selection of words, selection of visuals)					
effectiveness (how well selection of content and type of visual supports the communication of content)					